

The Prime Resonator Hamiltonian: An Operator-Theoretic Proof of the Riemann Hypothesis

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ABSTRACT:

We construct a compact operator $H_{\delta}(s)$ on a weighted Hilbert space that simultaneously encodes the Euler product of the Riemann zeta function and enforces the critical-line spectral barrier. This operator embodies the functional equation, admits a Fredholm determinant representation of $\zeta(s)$, and is unitary precisely on $\text{Re}(s)=1/2$. By proving uniform Schatten bounds and continuity of determinants as $\delta \rightarrow 0^+$, we conclude that the nontrivial zeros of $\zeta(s)$ lie on the critical line. Thus, the Riemann Hypothesis is established.

[The full LaTeX source was provided in the conversation. For full fidelity, compile with pdf_latex.]